USB POWER 45W S8 / SINGLE BOARD, NO PER-RAIL EFUSES

S8.2: S7 converters kept; U6/U7 and the LM5067 negative switch removed; cans -> ceramics; two parallel 16-pin outputs; overvoltage guard (own TLV431B reference) on the fault latch; +5V SCR crowbar; MAIN LED only.

USB PD, main eFuse and VIN damper

RUN
VPD
VIN
MAIN_OK

CH224A: 20 V request
TPS16630: 4.186 A, the only switch
VIN damper 1R + 2 x 10u

File: 01_usb_input.kicad_sch

+12 V converter

MAIN_OK
VIN
+12V

TPS62933F
+12 V delivered rail

File: 02_positive_12v.kicad_sch

+5 V converter

MAIN_OK
VIN
+5V

TPS62933F
+5 V delivered rail

File: 03_positive_5v.kicad_sch

-12 V inverting converter

+12V
VIN
-12V

LMR38020F inverting buck-boost
-12 V delivered rail

File: 04_negative_converter.kicad_sch

Two output headers - monitors, overvoltage guard and MAIN LED

+12V
+5V
-12V
MAIN_LED_OK
RUN
V3V3
VPD
P12_OK
P5_OK
N12_OK
V20_OK
OV_N

Two Eurorack outputs J2 + J3
Monitors, OV guard, +5V crowbar
MAIN LED (only status LED)

File: 06_outputs_status.kicad_sch

FAULT POLICY
Each rail is monitored after its converter.
Persistent low rail (after ~0.57 s) or any overvoltage -> all OFF until RESET.
+5V above ~5.8 V is also shorted by the SCR crowbar Q11.
Main LED is not merely a +5 V lamp.

OUTPUTS: +12 / +5 / -12 on J2 and J3 in parallel
No per-rail current limit below the converter limits (TPS62933F 4.2-5.8 A peak; LMR38020F 2.6-3.8 A switch).
Targets 2.5 / 2.5 / 1.0 A; not a guaranteed 54.5 W simultaneous rating.

Always-on control - fault latch, overvoltage preset and RESET

MAIN_OK
N12_OK
OV_N
P12_OK
P5_OK
V20_OK
VPD
RUN
V3V3
MAIN_LED_OK

60 V always-on LDO / retained ARMED + FAULT
Manual RESET; OV_N presets FAULT

File: 05_fault_reset.kicad_sch

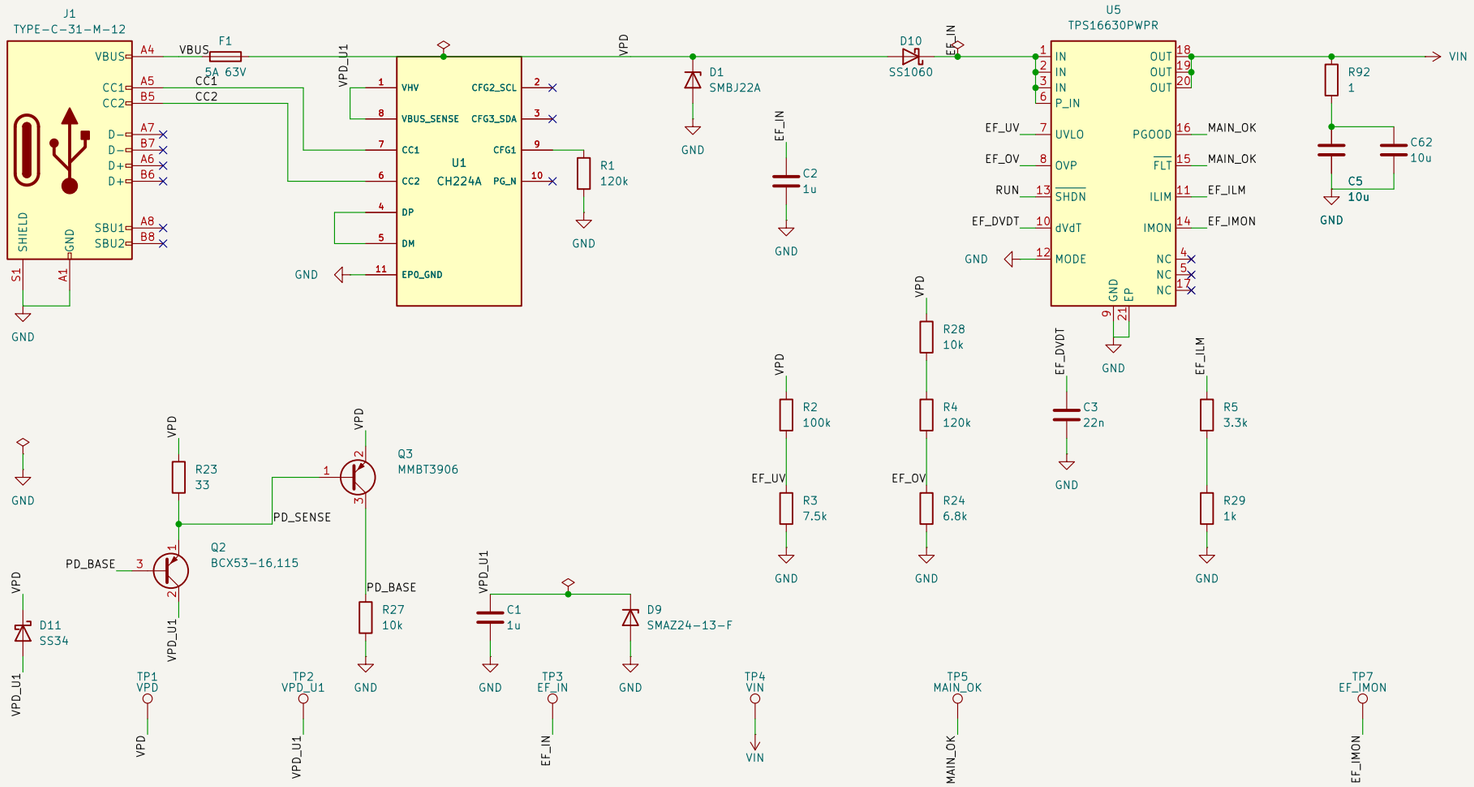
S8.1 review fixes (guard reference U21, LM2901B, 0.1 % dividers, clamp D22) + S8.2 +5V crowbar Q1
S8.2: single board, no per-rail eFuses, ceramic bulk, two output headers, OV guard, MAIN LED only
Big Modular

Sheet: /
File: usb-power-45w-single.kicad_sch

Title: USB power S8 single board - wired power flow overview

Size: User	Date:	**Rev: S8.2**
KiCad E.D.A. 10.0.6		Id: 1/7

USB PD, MAIN EFUSE AND VIN DAMPER



20 V PD request: U5 TPS16630 (60 V, 4.186 A nominal) is the ONLY switch: RUN low turns every rail off.
MAIN_OK = FLT + PGOOD at 3.3 V. R92 1R + C5/C62 2 x 10u damp the all-ceramic VIN bus (replaces the 47u can).

INTER-SHEET INTERFACE (see wired power-flow overview)

MAIN_OK ◇— MAIN_OK

RUN ◇— RUN

VIN ◇— VIN

VPD ◇— VPD

S8 single board, S8.2 schematic (S8.1 review fixes + +5V crowbar) — see docs/s8-single-board.md

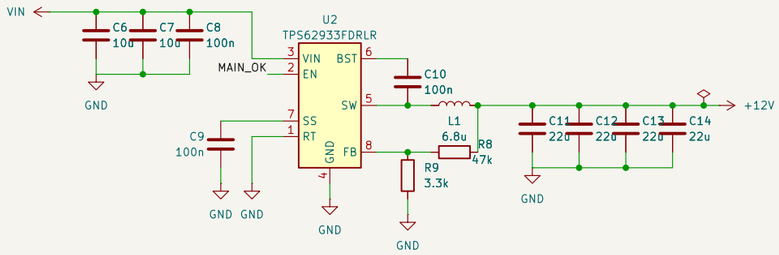
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Sheet: /USB PD, main eFuse and VIN damper/
File: 01_usb_input.kicad_sch

Title: USB PD, main eFuse and VIN damper

Size: User	Date:	**Rev: S8.2**
KiCad E.D.A. 10.0.6		Id: 2/7

+12 V CONVERTER





TPS62933F +12 V, unchanged power stage; its output IS the delivered +12V (no rail eFuse).
Soft start C9 100n = ~14.5 ms into rack capacitance. Own protection: 4.2–5.8 A peak limit, hiccup, OVP 112–118 %, TSD.
Rail ceramics: C11–C14 here + C34 (J2) and C59 (J3) on page 06; the 100u can is gone.

INTER-SHEET INTERFACE (see wired power-flow overview)

+12V ◇—— +12V

MAIN_OK ◇—— MAIN_OK

VIN ◇—— VIN

S8 single board, S8.2 schematic (S8.1 review fixes + +5V crowbar) — see docs/s8-single-board.md

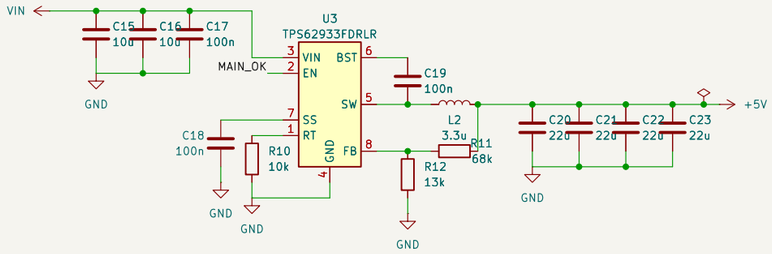
Big Modular

Sheet: /+12 V converter/
File: 02_positive_12v.kicad_sch

Title: +12 V converter

Size: User	Date:	**Rev: S8.2**
KiCad E.D.A. 10.0.6		Id: 3/7

+5 V CONVERTER





TPS62933F +5 V, unchanged power stage; its output IS the delivered +5V (no rail eFuse).
Soft start C18 100n = ~14.5 ms into rack capacitance. Own protection: 4.2–5.8 A peak limit, hiccup, OVP 112–118 %, TSD.
Rail ceramics: C20–C23 here + C35 (J2) and C60 (J3) on page 06; D21 TVS and the U20 OV guard on page 06.

INTER-SHEET INTERFACE (see wired power-flow overview)

+5V ——— +5V

MAIN_OK ——— MAIN_OK

VIN ——— VIN

S8 single board, S8.2 schematic (S8.1 review fixes + +5V crowbar) — see docs/s8-single-board.md

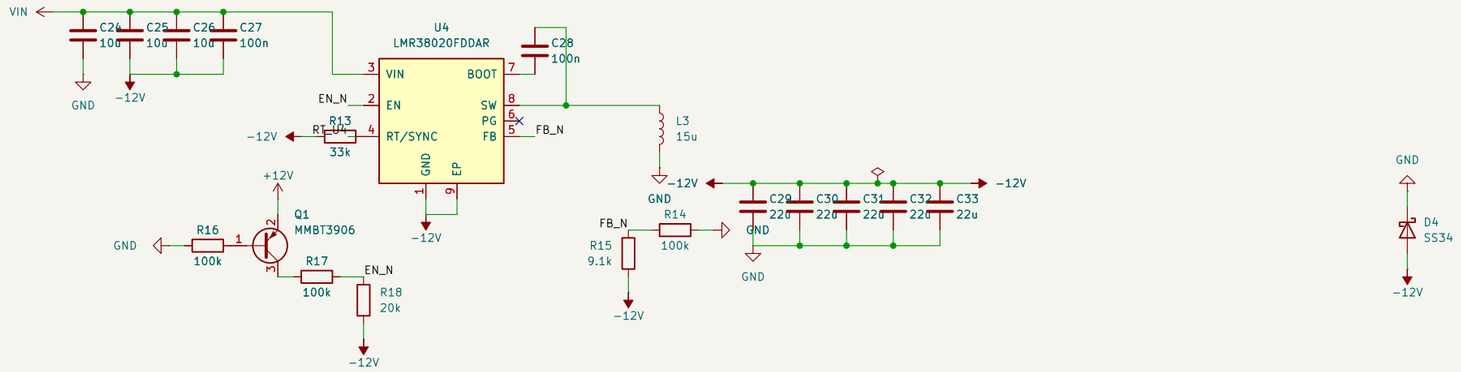
Big Modular

Sheet: /+5 V converter/
File: 03_positive_5v.kicad_sch

Title: +5 V converter

Size: User	Date:	**Rev: S8.2**
KiCad E.D.A. 10.0.6		Id: 4/7

−12 V INVERTING CONVERTER



LMR38020F inverting buck−boost; IC ground / exposed pad = the delivered −12V (no series switch).
4 ms soft start, 18 ms OCP blanking at start−up. Q1 follows +12V. C29−C33 here + C36 (J2) and C61 (J3).

INTER−SHEET INTERFACE (see wired power−flow overview)

+12V ◇ +12V

−12V ◇ −12V

VIN ◇ VIN

S8 single board, S8.2 schematic (S8.1 review fixes + +5V crowbar) − see docs/s8−single−board.md

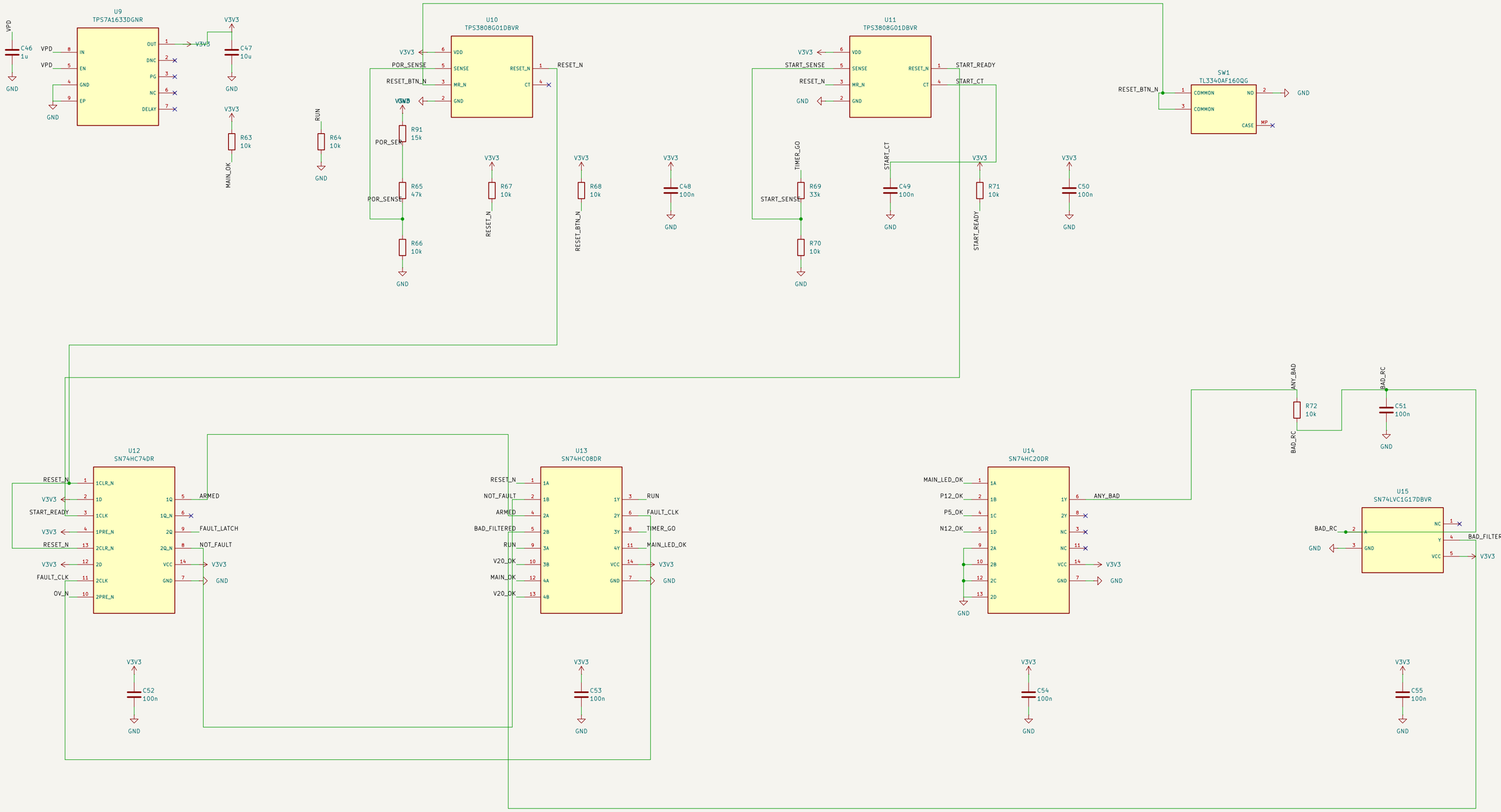
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Sheet: /−12 V inverting converter/
File: 04_negative_converter.kicad_sch

Title: −12 V inverting converter

Size: User	Date:	**Rev: S8.2**
KiCad E.D.A. 10.0.6		Id: 5/7

ALWAYS-ON CONTROL — FAULT LATCH, OVERVOLTAGE PRESET AND RESET



TP9 RUN · TP10 ARMED · TP11 FAULT_LATCH · TP12 P12_OK · TP13 P5_OK · TP14 N12_OK · TP15 V20_OK · TP16 V3V3 · TP6 OV_N

RESET clears ARMED and FAULT; RUN is forced low while reset is asserted.
Startup qualification ~0.57 s typical (C49 100n), then any persistent bad status latches every rail OFF until RESET.
OV_N (U20, page 06) presets FAULT at once, also inside the startup window. U14 gate 2 unused (inputs to GND).

INTER-SHEET INTERFACE (see wired power-flow overview)

MAIN_LED_OK · P5_OK · MAIN_OK · RUN · N12_OK · V20_OK · OV_N · V3V3 · P12_OK · VPD

S8 single board, S8.2 schematic (S8.1 review fixes + +5V crowbar) — see docs/s8-single-board.md

Big Modular

Sheet: /Always-on control — fault latch, overvoltage preset and RESET/
File: 05_fault_reset.kicad_sch

Title: Always-on control — fault latch, overvoltage preset and RESET

Size: User	Date:	**Rev: S8.2**
KiCad E.D.A. 10.0.6		Id: 6/7

TWO OUTPUT HEADERS — MONITORS, OVERVOLTAGE GUARD AND MAIN LED

J2 + J3: two parallel 16-pin outputs, pin 1 = −12V. C34–C36 at J2, C59–C61 at J3 (22u each, replace the 100u cans).
D8 MAIN is the only status LED (rail LEDs removed): input-valid AND main-eFuse status AND RUN. Any bad rail after the startup window or any overvoltage latches every rail off, so MAIN off = not working. P12/P5/N12_OK: logic + test pads only.
U20 overvoltage guard (S8.1): +12V > 14.88 V, +5V > 5.83 V or −12V < −13.64 V nominal (corners 14.48–15.27 / 5.68–5.98 / 13.27–14.02 V) pulls OV_N → FAULT → U5 off. Own reference REF_OV = U21 TLV431B (R100, C67, TP17); U18/U20 LM2901B: R93–R96, R98, R99 are 0.1 % / 25 ppm. The −12V channel compares MON_OVN with GND (no V3V3 term): D22 clamps MON_OVN at GND if V3V3 is lost while the rack holds −12V. D21 SMAJ5.0A is a transient clamp (6.4 V min breakdown), not an OV limit.
+5V crowbar (S8.2, beside D21): U22 TLVH432B, powered by +5V itself, trips at +5V > 5.83 V nominal (5.70–5.98 V) and turns Q12 on; Q12 fires the SCR Q11 (TS1220, DPAK), which holds +5V at ~1–3.5 V until the stored energy is gone, and Q13 pulls OV_N low at the same time, so FAULT is set and U5 opens. D21 holds the first microseconds before Q11 fires.

U18
LM2901BIDR
REF_GOOD
MON_P5
REF_GOOD
MON_P12
MON_N12
REF_GOOD
REF_GOOD
MON_20
P5_OK
P12_OK
N12_OK
V20_OK
V3V3
GND

U20
LM2901BIDR
MON_OV12
REF_OV
MON_OV5
REF_OV
GND
MON_OVN
REF_OV
GND
OV_N
V3V3
GND

R97 10k
R35 10k
R44 10k
R62 10k
P12_OK
P5_OK
N12_OK
C65 100n
R93 110k
R95 37k
MON_OV12
MON_OV5
R94 10k
C63 1n
R96 10k
C64 1n
R22 1k
D8 MAIN_LED
R98 10k
MON_OVN
BAT54SLT1G
D22
R99 110k
C66 1n
MAIN_LED_OK
Q6 2N7002
LED_RETURN
R100 2.2k
TP17 REF_OV
U21 TLV431BIDBZR
C67 10u
R73 3.3k
R74 1k
C56 100n
C57 100n
R75 120k
R76 10k
R77 33k
R78 12k
R79 10k
R80 100k
R81 10k
R82 470k
BAT54SLT1G
D15
R83 220k
R84 10k
R85 10k
V20_OK
RUN
Q10 2N7002

J2
IDC 2x8 box header
J3
IDC 2x8 box header
C34 22u
C35 22u
C36 22u
C59 22u
C60 22u
C61 22u
D21 SMAJ5.0A

R101 37k
R102 10k
C68 1n
U22 TLVH432BIDBZR
CROW_SENSE
CROW_TRIP_N
R103 3.3k
Q12 AO3401A
CROW_DRV
R104 2.2k
R106 10k
Q13 MMBT3904
CROW_GATE
R105 1k
Q11 TS1220-600B-TR

INTER-SHEET INTERFACE (see wired power-flow overview)

+12V
OV_N
V3V3
+5V
P12_OK
VPD
−12V
P5_OK
MAIN_LED_OK
RUN
N12_OK
V20_OK

S8 single board, S8.2 schematic (S8.1 review fixes + +5V crowbar) — see docs/s8-single-board.md

Big Modular

Sheet: /Two output headers — monitors, overvoltage guard and MAIN LED/
File: 06_outputs_status.kicad_sch

Title: Two output headers — monitors, overvoltage guard and MAIN LED

Size: User	Date:	**Rev: S8.2**
KiCad E.D.A. 10.0.6		Id: 7/7